

Md Tanvir Rouf Shawon

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 shawon-tanvir.github.io/  Scholar Profile

Profile

- 3rd-year PhD student in Computer Science at George Mason University specializing in Large Language Models, medical NLP, health informatics, and explainable AI. Experienced in multi-agent simulation frameworks and evaluation systems for healthcare AI. Passionate about building safe and reliable AI for real clinical use.
- Published 22 peer-reviewed research papers, including 3 Q1 journal articles, 1 Q2 journal article, and an EACL workshop paper.

Research Interests

Natural Language Processing, Health Informatics, Large Language Models, Machine Learning Application, Explainable Artificial Intelligence.

Education

George Mason University
Ph.D. in Computer and Information Sciences

Fairfax, Virginia, USA
August 2024 - Current

Ahsanullah University of Science and Technology
B. Sc. in Computer Science and Engineering

Dhaka, Bangladesh
2016 - 2021

Notre Dame Colege, Dhaka
Higher Secondary Certificate

Dhaka, Bangladesh
2013 - 2015

Selected Publications

(* denotes equal contribution)

- **Md. Tanvir Rouf Shawon**, Mohammad Sabik Irbaz, Hadeel R. A. Elyazori, Keerti Reddy Resapu, Yili Lin, Vladimir Franzuela Cardenas, K. Pierre Eklou, Farrokh Alemi, Kevin Lybarger "A Patient Simulation Framework for Risk Assessment of Conversational Health Care AI: Development and Evaluation Study" in *Journal of Medical Internet Research: Artificial Intelligence, JMIR*, 2026
- **Md. Tanvir Rouf Shawon**, G. M. Shahariar Shibli, Farzad Ahmed & Sajib Kumar Saha Joy "Explainable cost-sensitive deep neural networks for brain tumor detection from brain MRI images considering data imbalance" in *Multimedia Tools and Applications, Springer*, 2025
- G. M. Shahariar Shibli*, **Md. Tanvir Rouf Shawon***, Faisal Muhammad Shah, Mohammad Shafiul Alam & Md. Shahriar Mahbub "Bengali Fake Reviews: A Benchmark Dataset and Detection System" in *Neurocomputing, Elsevier*, 2024
- Kazi Elahi, Tasnuva Rahman, Shakil Shahriar, Samir Sarker, **Md. Tanvir Rouf Shawon**, and G. M. Shibli "A Comparative Analysis of Noise Reduction Methods in Sentiment Analysis on Noisy Bangla Texts" in *9th Workshop on Noisy and User-generated Text (W-NUT 2024), EACL*, 2024
- G. M. Shahariar Shibli, **Md. Tanvir Rouf Shawon**, Anik Hassan Nibir, Md. Zayed Miandad & Nibir Chandra Mandal "Automatic back transliteration of Romanized Bengali (Banglish) to Bengali" in *Iran Journal of Computer Science, Springer*, 2023
- **Md. Tanvir Rouf Shawon**, G. M. Shahariar Shibli, Faisal Muhammad Shah, Mohammad Shafiul Alam & Md. Shahriar Mahbub "Bengali Fake Review Detection using Semi-supervised Generative Adversarial Networks" in *5th International Conference on Natural Language Processing (ICNLP), Guangzhou, China, IEEE*, 2023
- Raihan Tanvir*, **Md Tanvir Rouf Shawon***, Md Humaion Kabir Mehedi, Md Motahar Mahtab & Annajiat Alim Rasel "A GAN-BERT Based Approach for Bengali Text Classification with a Few Labeled Examples" in *Distributed Computing and Artificial Intelligence, 19th International Conference, L'Aquila, Italy, Springer*, 2022

- **MD Tanvir Rouf Shawon***, Raihan Tanvir*, Humaira Ferdous Shifa, Susmoy Kar & Mohammad Imrul Jubair **"Jamdani Motif Generation using Conditional GAN"** in *23rd International Conference on Computer and Information Technology (ICCIT)*, Dhaka, Bangladesh, IEEE, 2020

Research Experience

Graduate Research Assistant (Summer) Development of an Advanced AI System for Depression Management. PCORI-funded project — PIs: Dr. Farrokh Alemi (College of Public Health) & Dr. Kevin Lybarger (College of Engineering and Computing), GMU
July 2025 - August 2025; May 2026 - August 2026

(CO-PI) Bengali Fake Review: A Benchmark Dataset and Detection System. Internal Research Grant supported by Committee for Advance Studies and Research (CASR), AUST
Jan 2022 - Jan 2023

Teaching Experience

George Mason University (GMU)

Graduate Teaching Assistant

Aug 2025 - Present

Department of Computer Science

Courses: Database Concepts, Introduction to Low-Level Programming

Ahsanullah University of Science and Technology (AUST)

Lecturer

June 2021 - July 2024

Department of Computer Science & Engineering

Courses: Soft Computing, Pattern Recognition, Algorithms, Object Oriented Programming

Conference Presentation

- Bengali Fake Review Detection using Semi-supervised Generative Adversarial Networks, *5th International Conference on Natural Language Processing (ICNLP)*, Guangzhou, China. [Presentation link: <https://youtu.be/rd5N6z5-H0U>] March 24-26, 2023
- Effectiveness of Transformer Models on IoT Security Detection in StackOverflow Discussions, *1st International Conference on Information and Communication Technology for Development*, Khulna, Bangladesh. [Presentation link: <https://youtu.be/Y4ntugjHEGM>] July 29-30, 2022

Awards & Achievements

- Dean's List of Honor, AUST 2021
- Government Scholarship 2010, 2013
- I-Genius, Grameenphone Prothom Alo Internet Festival 2013

Skills

- **Programming Language:** Python, Java, JavaScript, C, C++, PHP, HTML, CSS
- **ML Tools:** PyTorch, Keras, Tensorflow, Scikit-learn, GANs, RAG
- **NLP:** Hugging Face, LangGraph, LangChain, GPT; **Visualization:** Matplotlib, Seaborn
- **Database:** MS SQL, MySQL, SQLite, Oracle PL SQL

Professional Activities and Services

- Reviewer, Frontiers in Artificial Intelligence
- Reviewer, International Journal of Image and Graphics
- Reviewer, International Journal of Informatics and Communication Technology (IJ-ICT)
- Co-opt member (Extra Curricular Activities) - AUST CSE SOCIETY
- Advisor of Cefalo CodeFiesta 2022: AUST Inter-University Programming Contest